MICROECONOMICS OF COMPETITIVENESS · GROUP 1

Energy Cluster Norway

From inherited to created prosperity

Raphael Knispel · Daniel Mitterlehner · Tim Orler · Elias Johannes Schuchnigg · Márton Szakasits-Doroszlói · Filip Marko Župaric

FH Wien der WKW · Executive Management MA2 · WS 2026/27 · Dr. Clemens Löffler, Dr. Kinga Niemczak

CENTRAL THESIS

Norway must turn the prosperity it *inherited* from petroleum into prosperity it creates with globally competitive energy technologies.

RESEARCH QUESTION

How can Energy Cluster Norway help turn world-class offshore capabilities into globally competitive energy technologies?

1 · Diagnosis

Performance, business environment and policy

2 · Cluster

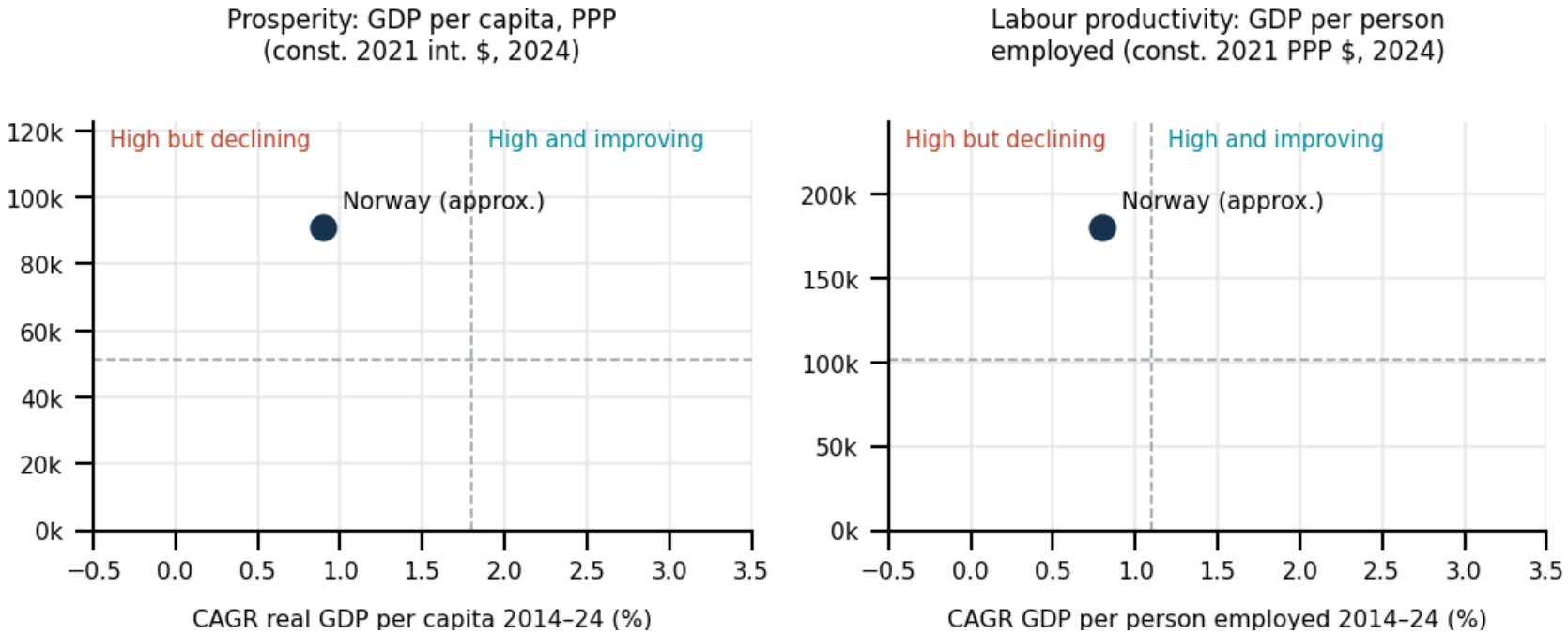
History, size, actors and Diamond

3 · Strategy

Value proposition, ECN pipeline, recommendations, limits

ECONOMIC PERFORMANCE · MOC QUADRANTS

High level, but falling behind on growth



MOC QUADRANT: HIGH BUT DECLINING

91k USD per capita · **0.9%** growth p.a. (avg 1.8%) · **100.3** USD/hour, 3rd in OECD · **57%** of goods exports petroleum

Sources: Löffler & Niemczak (2026), MOC exhibits (World Bank / ILO; positions approximate); OECD (2026); Norwegian Petroleum (2026). Figures include the petroleum sector.

ECONOMIC PERFORMANCE · MACRO AND MICRO · ORANGE = WATCH POINT

Strong buffers, weak commercial output

MACRO PERFORMANCE

GDP growth	Mainland +1.7% in 2025; about +1.7% expected for 2026
Inflation	Converging towards the 2% target by 2027
Currency	Floating krone; a weak krone is no substitute for productivity
Employment	Unemployment around 4%; participation declining
Fiscal stability	2.7% of the oil fund spent in 2026; rule is 3%
Non-oil deficit	12.6% of mainland trend GDP: leans on the fund

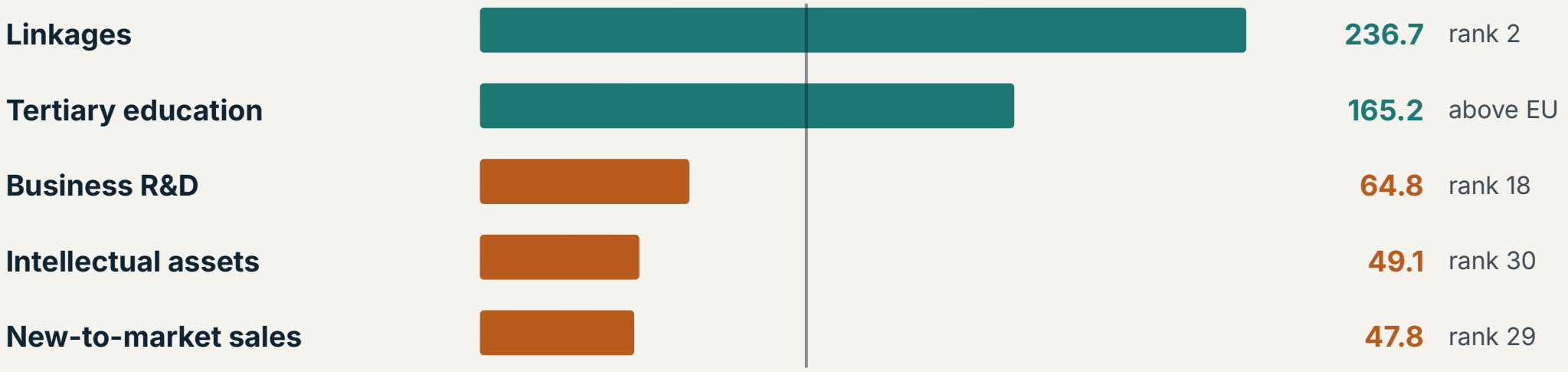
MICRO PERFORMANCE

Trade	57% of goods exports are petroleum; suppliers export ≈ NOK 200bn
FDI	Net inflows ≈ USD 8bn (2023); volatile year to year
Education	Tertiary index at 165% of the EU average
Patents, R&D	IP assets rank 30 of 40; business R&D 65% of EU
Clusters	Mature NIC programme; ECN unites three clusters, 350+ members
Survey data	WEF GCI 2019: rank 17; institutions 8th, social capital 3rd

Sources: Statistics Norway; Ministry of Finance; IMF; Norwegian Petroleum; Norwep; World Bank; EC (2026b); WEF (2019).

INNOVATION PERFORMANCE · EU AVERAGE = 100 (VERTICAL LINE)

Strong in collaboration, weak in commercial results



> Knowledge is created and shared, but rarely converted into protected assets, new products and high-tech exports (rank 38 of 40).

Source: European Commission (2026b), European Innovation Scoreboard 2026, country profile Norway. Overall: Strong Innovator, 115.7% of EU, rank 10. Output shares are depressed by the petroleum-heavy export base.

BUSINESS ENVIRONMENT · LEGACY, REGION, MACRO POLICY

Legacy, geography and macro policy

LEGACY: WHAT NORWAY INHERITED

Culture, religion. Lutheran, egalitarian, high-trust; communal work (dugnad).

Language, society. Small population (5.6m), one language, Nordic culture.

Institutions. Nordic model: rule of law, state-employer-union cooperation, 67% state share in Equinor.

Resources. Hydropower, shipping, fisheries; oil since Ekofisk (1969).

REGIONAL POSITION AND INTERACTION

Geography. Remote, high-cost north-western periphery; harsh North Sea.

Market access. Outside the EU, but in the EEA (1994), Schengen and NATO.

Energy links. Gas supplier to Europe; CO_2 storage for NL, DK and SE.

Neighbours. UK, DE, NL, DK: customers, partners and rivals.

MACRO POLICY: A STRENGTH, NOT A CONSTRAINT

Monetary policy

2% inflation target and floating krone since 2001; inflation expected back at target by 2027.

Fiscal rule, stabilisation

Spend only the fund's expected real return (3% since 2017; 2.7% in 2026), with room for countercyclical spending.

Currency, wealth buffer

The krone weakens when oil falls, helping non-oil exporters; the fund keeps oil income out of the economy.

Sources: Own assessment; IMF (2025); Ministry of Finance (2026a, 2026b); Equinor (2026). EEA = European Economic Area.

BUSINESS ENVIRONMENT · MICRO POLICY AND DIAMOND

Micro policy and the national Diamond

POLICY AREA	CHOICE	RATING
Structural, liberalisation	EEA market access; open trade; 78% petroleum tax rate.	**++ open**
Competition	Open licensing rounds, but a few large buyers dominate.	**± concentrated**
Privatisation	Equinor listed in 2001; state keeps 67%.	**± state-led**
Education, R&D	Tertiary education at 165% of EU; SkatteFUNN R&D tax credit.	**+ strong**
Cluster policy	Norwegian Innovation Clusters; regional bodies merged into ECN.	**+ in place**
Demand policy	State funds Longship, Hywind Tampen (via Enova) and Utsira Nord.	**± state-funded**

NATIONAL DIAMOND

Factor conditions ++

Educated workforce, hydropower, capital; high wages.

Demand +

Sophisticated buyers, strict HSE; small home market (5.6m).

Related industries ++

Dense supplier, maritime and research networks; regionally fragmented.

Strategy and rivalry ±

High trust, but few large firms; low business R&D.

Rich but narrow: slow to create new industries.

Sources: Innovation Norway (n.d.); Equinor (2026); Ministry of Energy (2026); EC (2026b); Porter (2008). ++ strong, + good, ± mixed.

CLUSTER HISTORY

Five phases: from discovery to energy transition

1960s–70s

Foundations

Ekofisk (1969) starts production in 1971; the state creates Statoil and the Petroleum Directorate (1972) and uses local-content rules to open the shelf to Norwegian firms.

1980s–90s

Capability building

Yards and engineers such as Aker and Kværner learn offshore work; the NORSOK programme (1993) cuts project costs; harsh-environment skills mature.

2000s

Going global

Subsea suppliers such as Aker Solutions, Subsea 7 and Kongsberg win international markets; Statoil is listed in 2001; regional cluster bodies emerge.

2014–20

Shock and reset

The 2014 oil-price fall forces cost cuts and layoffs; suppliers look to offshore wind and CCS; the Hywind pilot and Longship show new paths.

2021–26

Transition and consolidation

Hywind Tampen (2022/23) and the first CO_2 stored by Northern Lights (2025); ECN launches in August 2026 as one national platform.

PATTERN Each phase added a capability; collaboration institutions came last.

Sources: Norwegian Petroleum (2026); Equinor (n.d.-a, n.d.-b); Northern Lights JV (2026). Phases are our own grouping.

CLUSTER MAP

Offshore energy technology and services

SPECIALISED SUPPLIERS

Aker Solutions, TechnipFMC, Subsea7, Kongsberg; vessels, digital and AI services, SMEs

KNOWLEDGE AND TALENT

NTNU, UiB, UiS, NHH, UiO; SINTEF, NORCE, IFE; test sites such as METCentre

Core cluster

Oil and gas · floating and fixed offshore wind · CO_2 transport and storage · offshore electrification · decommissioning

Operators: Equinor (67% state), Aker BP, Vår Energi. Flagships: Hywind Tampen, Northern Lights

DEMAND

Shelf operators; European emitters buying CO_2 storage; utilities and industry

RELATED CLUSTERS

Maritime, hydropower and grid, process industry, aquaculture, ICT

IFCS **Energy Cluster Norway** (GCE Ocean Technology + Energy Transition Norway + Energy Valley) · Offshore Norge · Norwegian Offshore Wind · Norwep · Innovation Norway. Government and regulators: ministries, Havtil, Gassnova, Enova

Own illustration in the format of Porter's cluster maps (Porter, 2008); selection of actors illustrative, not exhaustive.

CLUSTER SIZE AND PERFORMANCE

A large, export-oriented cluster in transition

NOK 200bn

energy-supplier exports per year (approx.)

350+

member companies in ECN

67%

state share in Equinor, the largest buyer

1.5 → 5+ Mt

Northern Lights CO_2 capacity, phase 1 → phase 2

[add]

employees in the supplier industry, 2014 → latest year

[add]

share of supplier revenue from wind, CCS, electrification

READING Strong export base; the question is how fast new segments grow.

Sources: Norwegian Energy Partners (2026); Energy Cluster Norway (2026); Equinor (2026, n.d.-b). [add]: supplier-industry statistics, see report Section 5.2.

ENERGY CLUSTER NORWAY

One national platform, three equal regions

Launched August 2026 | **350+ members** | **Equal ownership: ⅓ per region** | **Fully integrated by end-2027**

Bergen / Vestland

GCE Ocean Technology

Subsea and ocean technology, CO_2 terminal at Øygarden: deep tech and scale-up

Stavanger / Rogaland

Energy Transition Norway

Operators, supplier networks, Utsira Nord: demand and first customers

Oslo region

Energy Valley

Headquarters, finance, digitalisation and AI: capital and internationalisation

FROM OUR CEO INTERVIEW

Model built bottom-up from interviews with 100+ companies. Services are distributed, not duplicated: MIT scale-up programme in Bergen and Boston, Technical Committee in Stavanger, open to all members. Next milestone: a secure funding model by spring 2027.

Sources: Energy Cluster Norway (2026); Energy Valley (2026); O. Hagesæther, CEO of ECN, written interview, 25 October 2026.

CLUSTER DIAMOND

Strong inputs, weak demand for new technology at scale

Factor conditions — ++ · stable

World-class subsea engineering, test sites and research; scarce specialists; grid limits

Demand — + · CCS rising

Very sophisticated in oil and gas; for CCS and floating wind still reliant on state support

Related industries — ++ · rising

Maritime, power and process industries recombine; ECN reduces regional fragmentation

Strategy and rivalry — ± · stable

Few powerful buyers (Equinor 67% state-owned); low business R&D; scale-ups stay project-based

THE WEAK LINK

Buyer concentration and state-backed projects decide whether a demonstration ever becomes a commercial order.

Own assessment. Strength: ++ strong, + good, ± mixed. Based on European Commission (2026b), Equinor (2026), Bjørndal et al. (2025).

STRATEGIC ISSUES

Strategic issues and policies already in place

ISSUE	POLICY IN PLACE	REMAINING GAP
Demand Few buyers; pilots rarely become orders.	State aid: Utsira Nord (cap NOK 35bn), Enova support for Hywind Tampen, Longship CCS.	**No lead customer or procurement route built into support.**
Strategy and rivalry Operator concentration; business R&D at 65% of EU.	SkatteFUNN tax credit; open licensing rounds; state ownership of Equinor.	**Incentives reward R&D spend, not commercial conversion.**
Factor conditions Scarce specialists, grid limits, IP output (rank 30).	Research Council programmes (e.g. Green Platform); universities; test sites such as METCentre.	**Test infrastructure and skills not coordinated across regions.**
Related clusters and IFCs Regional fragmentation of cluster bodies.	Norwegian Innovation Clusters programme; ECN merger in 2026.	**No common gate process or outcome metrics yet.**

Sources: Own assessment; Innovation Norway (n.d.); Ministry of Energy (2026); Equinor (n.d.-b); European Commission (2026b).

EVIDENCE: TWO PILOT-TO-SCALE PATHS

Validation alone does not create scale

	Hywind · floating wind	Northern Lights · CO_2 storage
Pilot	Demonstrator off Karmøy, 2009	Longship CCS, investment decision 2020
First customer	Hywind Tampen powers Equinor's Snorre and Gullfaks fields (2022/23)	State-supported Norwegian emitters; first CO_2 stored 2025
Next step	Utsira Nord: state-aid competition, aid capped at NOK 35bn	Contracts in NL, DK and SE; phase 2 from 1.5 to 5+ Mt CO_2 p.a.
Lesson	Still policy-dependent: terms revised in 2026 over an immature supply chain	Anchor customers plus an export market turned a public project into a service

Both paths moved forward only once a committed buyer and a market route were in place.

Sources: Equinor (n.d.-a, n.d.-b); Ministry of Energy (2026); Northern Lights JV (2026); offshoreWIND.biz (2026).

POSITIONING · WHY NORWAY?

Three North Sea locations compete for the same role

Aberdeen / Scotland

Closest match to Norway's oil and gas supply chain; large floating-wind leasing (ScotWind); CCS plans in the North Sea.

Esbjerg / Denmark

Europe's leading port for installing offshore wind; Danish CO_2 storage projects in the North Sea.

Rotterdam / Netherlands

Europe's largest port and industrial cluster; CCS infrastructure for its own emitters.

NORWAY'S EDGE

Cross-border CO_2 storage in operation (Northern Lights).

Floating wind in harsh seas (Hywind Tampen).

Weak spot: higher costs, small manufacturing base.

IMPLICATION Compete on proven harsh-environment systems, not on volume.

Sources: Own assessment; Equinor (n.d.-a); Northern Lights JV (2026); competitor sources see report Section 8.1.

NATIONAL VALUE PROPOSITION

Norway as Europe’s reference location for offshore energy systems in harsh environments

The place where offshore energy technologies are proven, scaled and exported: from oil and gas to floating wind, CO_2 storage and offshore electrification.

WHAT NORWAY SHOULD NOT DO

No subsidy race for mass manufacturing without a cost or scale advantage

No duplicate test infrastructure across the three regions

No permanent operating subsidies without a credible path to competitiveness

STRATEGIC PRIORITIES FOR ECN

One national pipeline from industrial need to first order

Gate 1	Gate 2	Gate 3	Gate 4	Gate 5	Gate 6
Verified industrial need	Capable consortium	Technical validation	Demonstration	First commercial order	International scale-up

Before the demonstration gate: a named lead customer, purchasing criteria and a procurement route.

1 · Convert pilots into first commercial orders

Common six-gate process; measure conversion, not activity

2 · Build one national commercialisation pipeline

Shared gates and stop/continue decisions across regions; challenge roadmaps without pre-selected technologies

3 · Scale successful technologies internationally

Norwep, export finance and North Sea partners on board before the demonstration ends

WHO DECIDES?

Public funders link co-funding to the gates; ECN prepares the reviews with its Technical Committee.

WHY COMMIT EARLY?

The first unit is co-funded, risk is shared and purchasing criteria are set with suppliers.

WHO PAYS FOR ECN?

Gate reviews become a member service in the funding model due in spring 2027.

Stage-gate logic (Cooper, 1990) applied at cluster level; O. Hagesæther, written interview, 25 October 2026.

POLICY RECOMMENDATIONS

Recommendations: who needs to do what

National government

- Turn support into demand: a multi-year, technology-neutral procurement calendar for low-emission offshore energy.
- Tie aid to a named lead customer and a falling support path.

Regional and local government

- Fast-track ports, land and grid connections in Bergen, Stavanger and Øygarden.
- Share, not duplicate, test sites and skills programmes across the regions.

Private sector

- Operators act as lead customers and publish purchasing criteria for first units.
- Suppliers raise business R&D, protect IP and co-fund demonstrations.

Universities and institutes

- Reward IP, spin-outs and industrial PhDs, not only publications.
- Open test sites and train subsea, wind and CCS specialists.

Trade associations

- Bundle member demand into joint buyer roadmaps (Offshore Norge, Norwegian Offshore Wind).
- Lead export promotion with Norwep and Innovation Norway.

Collaboration bodies (ECN)

- Run the six-gate pipeline; funders link co-funding to the gates.
- Report conversion: pilot-to-order rate and export revenue.

Source: Own recommendations, derived from the Diamond analysis and the Hywind and Northern Lights cases.

IMPLEMENTATION AND SCORECARD

Measure conversion, not activity

0–12 months

Governance for 2027, challenge map, gate criteria, portfolio baseline

Proof: shared data model, baseline conversion data

12–36 months

Challenge-based calls, cross-regional consortia, lead customers, co-funded demonstrations

Proof: more projects with purchasing criteria and private co-finance

36–60 months

Scale what works, repeat procurement, target-market partnerships, stop weak projects

Proof: first deployments, repeat orders, export contracts

Pilots into orders
Share with named lead buyer; pilot-to-order rate

One pipeline
Time from need to pilot; private funding per public NOK

Scale-up
Repeat orders; export revenue; EIS output indicators

Shared value
Verified emission savings; supplier growth; skills

CREATING SHARED VALUE

Subsea CCS: Northern Lights sells CO_2 storage as a service: exports and decarbonisation at once.

Floating wind: Hywind Tampen powers oil and gas platforms: lifecycle services and lower emissions.

Sources: Own analysis based on Porter & Kramer (2011); Northern Lights JV (2026); Equinor (n.d.-a).

CONCLUSION

Norway has the knowledge and the collaboration. What it needs is a reliable path from pilot to first order.

Diagnosis

High but declining; a commercialisation gap

Bottleneck

Demand for new technology at commercial scale

Role of ECN

One pipeline, shared gates, outcome-based measurement

LIMITS

EIS output shares partly reflect the petroleum export base · two state-backed cases · one interview with ECN itself. The final report adds cluster statistics and a second interview.

Thank you. We look forward to your questions.